

T11

$$d) X_{\min} < C$$

$$\alpha_2 \rightarrow \min$$

$$\alpha_1 \leq \alpha$$

$$\alpha_1 = P(X_{\min} < C | H_0) \leq \alpha$$

$$P(X_{\min} < C) = 1 - P(X_{\min} \geq C) = 1 - P(X \geq C)^n =$$

$$= 1 - (1 - F(C))^n$$

Для равномерного распределения $F(C) = C$,

$$1 - (1 - C)^n \leq \alpha$$

$$(1 - C)^n \geq 1 - \alpha$$

$$1 - C \geq \sqrt[n]{1 - \alpha}$$

$$\underline{C \leq 1 - \sqrt[n]{1 - \alpha}}$$

$$W = P(X_{\min} < C | H_1) = 1 - (1 - \Phi(C))^n =$$

$$= 1 - \left(1 - \frac{e}{e-1} (1 - e^{-C})\right)^n \leq \Phi(C) = \int_0^C \frac{e^{1-x}}{e-1} dx = \frac{e}{e-1} (1 - e^{-C})$$

$$\alpha_2 = P(X_{\min} \geq C | H_1) = (P(X \geq C | H_1))^n = (1 - \Phi_1(C))^n \rightarrow \min$$

$\alpha_2 \rightarrow \min$, когда $F(C)$ max, т.к. $F(C)$ монотонно возрастает, то это выполняется при $C_{\max}$, т.е. при $C = 1 - \sqrt[n]{1 - \alpha}$

$$\alpha_1 = \left(1 - \frac{e}{e-1} (1 - e^{-(1 - \sqrt[n]{1 - \alpha})})\right)^n$$

$$G: X_{\min} < 1 - \sqrt[n]{1 - \alpha}$$

$$W = 1 - \left(1 - \frac{e}{e-1} (1 - e^{-(1 - \sqrt[n]{1 - \alpha})})\right)^n = 1 - \left(1 - \frac{e}{e-1} \left(1 - e^{-\left(1 - \frac{1}{n} \ln(1 - \alpha) + o\left(\frac{1}{n}\right)\right)}\right)\right)^n$$

$$= 1 - \left(1 - \frac{e}{e-1} \left(1 - \left(1 + \frac{1}{n} \ln(1 - \alpha) + o\left(\frac{1}{n}\right)\right)\right)\right)^n$$

$$= 1 - \left[1 + \frac{e}{e-1} \frac{1}{n} \ln(1 - \alpha) + o\left(\frac{1}{n}\right)\right]^n \xrightarrow{n \rightarrow \infty} 1 - e^{\frac{e}{e-1} \ln(1 - \alpha)} =$$

$$21 - (1 - \alpha) \frac{p}{p-1} \neq 1, \quad \text{крит. не сов.}$$

T12] $\alpha = 0,2$

$$H_0: \theta \sim p(\omega) = \frac{1}{3} \{4\} + \frac{1}{6} \{3\} + \frac{1}{4} \{2\} + \frac{1}{4} \{1\}$$

$$H_1: \theta \sim p(\kappa) = \frac{1}{4} \{1\} + \frac{1}{4} \{2\} + \frac{1}{4} \{3\} + \frac{1}{4} \{3\}$$

$n=2$ броска

Матрица отношений правдоподобий. ($\ell = \frac{4}{L_0}$)

отно на осн	1	2	3	4
1	1	1	$3/2$	$3/4$
2	1	1	$3/2$	$3/4$
3	$3/2$	$3/2$	$9/4$	$9/8$
4	$3/4$	$3/4$	$9/8$	$9/16$

$$G: \ell(\bar{X}) \geq c$$

$$P(\bar{X} \in G | H_0) \leq 0,2$$

берём макс знач ℓ

$$1) \ell_1 = 9/4 \quad (3,3)$$

$$L_0 = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36} = 0,028 < 0,2$$

$$L_1 = \frac{1}{16}$$

$$2) \ell_2 = 3/2 \quad [(1,3), (2,3), (3,1), (3,2)]$$

$$L_0 = 4 \cdot \left(\frac{1}{6} \cdot \frac{1}{6} \right) = \frac{1}{6} = 0,167$$

$$L_1 = 4 \cdot \frac{1}{16} = \frac{1}{4}$$

$$\alpha = 1) L_0 + 4) L_0 = \frac{1}{36} + \frac{1}{6} = \frac{2}{36} = 0,194 \quad (\text{это близко к } 0,2)$$

$$3) \ell_3 = 9/8 \quad [(3,4), (4,3)]$$

$$L_0 = 2 \cdot \left(\frac{1}{6} \cdot \frac{1}{3} \right) = \frac{1}{9} = 0,11$$

$$L_1 = 2 \cdot \frac{1}{16} = \frac{1}{8}$$

$$\alpha = 1) L_0 + 2) L_0 + 3) L_0 = 0,214 > 0,2$$

$$W = \sum_{i=1}^3 L_1 = \frac{1}{16} + \frac{1}{4} = \frac{5}{16}$$

$$G = \{(3,3), (1,3), (2,3), (3,1), (3,2)\}$$

Т13 $\alpha = 0,05$

$n = 139$

$m = 1000$

$H_0: \sigma_{1gn}^2 = \sigma_{2gn}^2$

$H_1: \sigma_{1gn}^2 \neq \sigma_{2gn}^2$

~~$\Delta = \frac{\sum_{i=1}^n \frac{x_i^2}{\sigma_i^2}}{\sum_{i=1}^n \frac{1}{\sigma_i^2}}$~~

$\Delta = \frac{\sum_{i=1}^n \frac{x_i^2}{\sigma_i^2}}{\sum_{i=1}^n \frac{1}{\sigma_i^2}}$

$\bar{\sigma}_{gn} = \frac{\sum_{i=1}^n \sigma_{gn}^2}{\sum_{i=1}^n 1} = 0,868$

$\bar{\sigma}_{шпр} = \frac{\sum_{i=1}^n \sigma_{шпр}^2}{\sum_{i=1}^n 1} = 0,8376$

$\sigma_{1gn} = 5,722$

$\sigma_{2gn} = 6,161$

$(\sigma_1^2 = \sigma_2^2)$

(аналогичн. м. показн. для ширины зеркала)

$\sum_{i=1}^n \sigma_i^2 \cdot \frac{n}{n-1}$

$\sum_{i=1}^n \sigma_{1gn}^2 = \frac{139}{138} \cdot (5,722)^2 = 32,9785$

$\sum_{i=1}^n \sigma_{2gn}^2 = \frac{1000}{999} \cdot (6,161)^2 = 37,9959$

$\sum_{i=1}^n \sigma_{шпр}^2 = \frac{139}{138} \cdot (4,612)^2 = 21,4246$

$\sum_{i=1}^n \sigma_{шпр}^2 = \frac{1000}{999} \cdot (5,055)^2 = 25,5786$

$\sigma_{шпр} = 4,612$

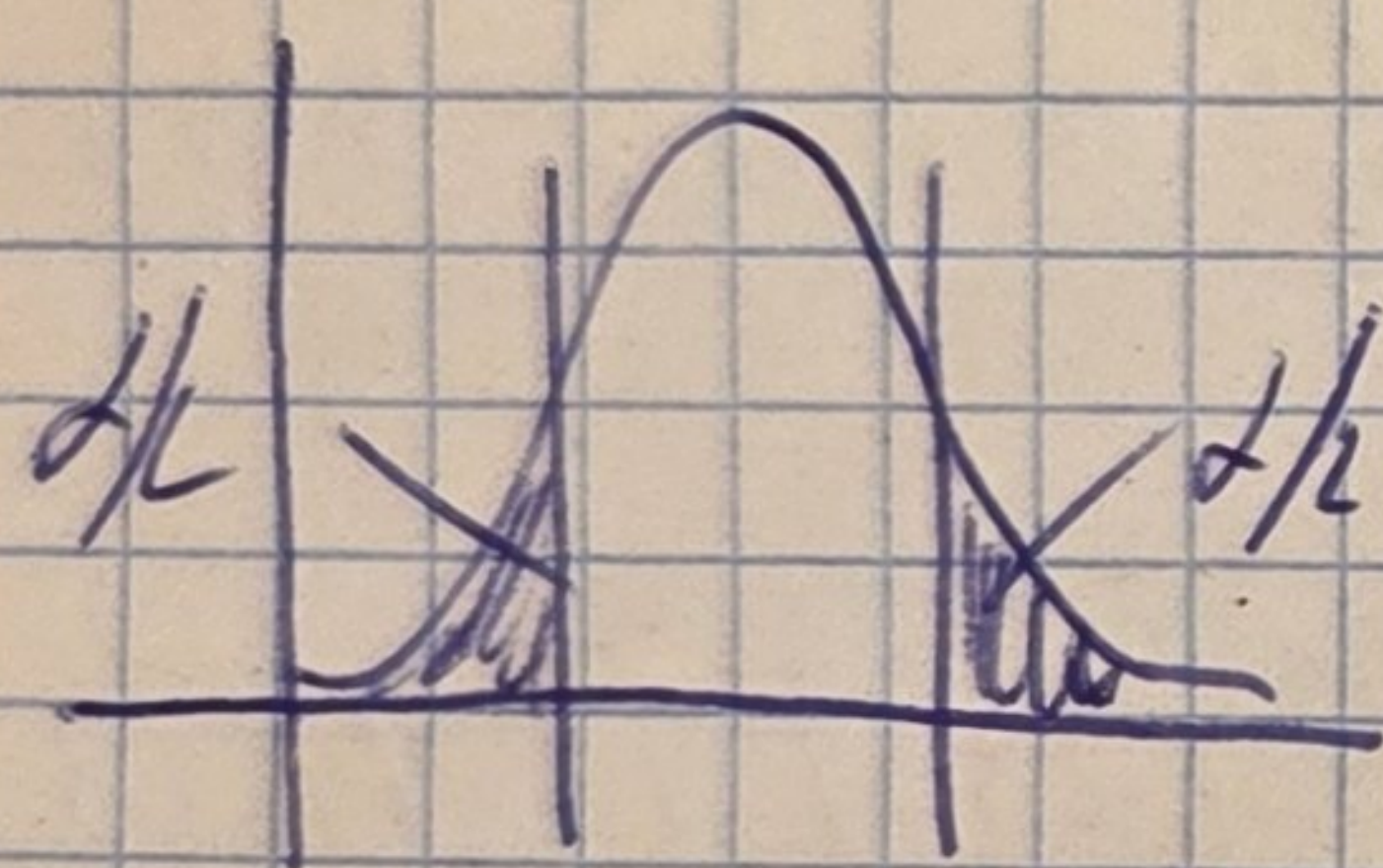
$\sigma_{шпр} = 5,055$

$\xi \sim N(\theta, \sigma_{\xi}^2)$

$\eta \sim N(\psi, \sigma_{\eta}^2)$

случ. расхожд

$\Delta \sim F(n-1, m-1) = F(138, 999)$



$G: \bar{\sigma} \leq F_{\alpha/2}(138, 999) \cup \bar{\sigma} \geq F_{1-\alpha/2}(138, 999)$

$t_1 = F_{\alpha/2}(138, 999) = 0,7674$

$t_2 = F_{1-\alpha/2}(138, 999) = 1,2717$

$G = (-\infty, 0,7674] \cup [1,2717, +\infty)$

$\bar{\sigma}_{gn} \text{ и } \bar{\sigma}_{шпр} \notin G \Rightarrow$ нет осн. ответа. Но (расх. случает)

$W(\theta) = P(\Delta \in G | H_1) = P(\Delta \leq t_1 | H_1) + P(\Delta \geq t_2 | H_1)$

$\Delta = \frac{\sum_{i=1}^n \frac{x_i^2}{\sigma_i^2}}{\sum_{i=1}^n \frac{1}{\sigma_i^2}} = \frac{\sum_{i=1}^n \frac{x_i^2}{\sigma_i^2}}{\sum_{i=1}^n \frac{1}{\sigma_i^2}} > \frac{\sigma_1^2}{\sigma_2^2} = \theta$

$W(\theta) = P\left(\frac{\sum_{i=1}^n \frac{x_i^2}{\sigma_i^2}}{\sum_{i=1}^n \frac{1}{\sigma_i^2}} \leq t_1\right) + P\left(\frac{\sum_{i=1}^n \frac{x_i^2}{\sigma_i^2}}{\sum_{i=1}^n \frac{1}{\sigma_i^2}} \geq t_2\right) = \int_{-\infty}^{t_1} g(t) dt + \int_{t_2}^{+\infty} g(t) dt$

$= P\left(\frac{\sum_{i=1}^n \frac{x_i^2}{\sigma_i^2}}{\sum_{i=1}^n \frac{1}{\sigma_i^2}} \leq \frac{t_1}{\theta}\right) + P\left(\frac{\sum_{i=1}^n \frac{x_i^2}{\sigma_i^2}}{\sum_{i=1}^n \frac{1}{\sigma_i^2}} \geq \frac{t_2}{\theta}\right) =$

$= \int_{\frac{t_1}{\theta}}^{+\infty} g(t) dt + 1 - \int_{\frac{t_2}{\theta}}^{+\infty} g(t) dt$

T14

$$\bar{x}_n, x = \{-1, 11, -6, 10, 242\}, n=3$$

$$q \sim N(a, \sigma_x^2), \sigma_x^2 = 2$$

$$\alpha = 0.05$$

$$\bar{y}_m, y = \{-2.29, -2.91\}, m=2$$

$$z \sim N(b, \sigma_y^2), \sigma_y^2 = 1$$

$$H_0: a = b$$

$$H_1: a > b$$

$$\bar{x} - \bar{y} \sim N(a - b, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m})$$

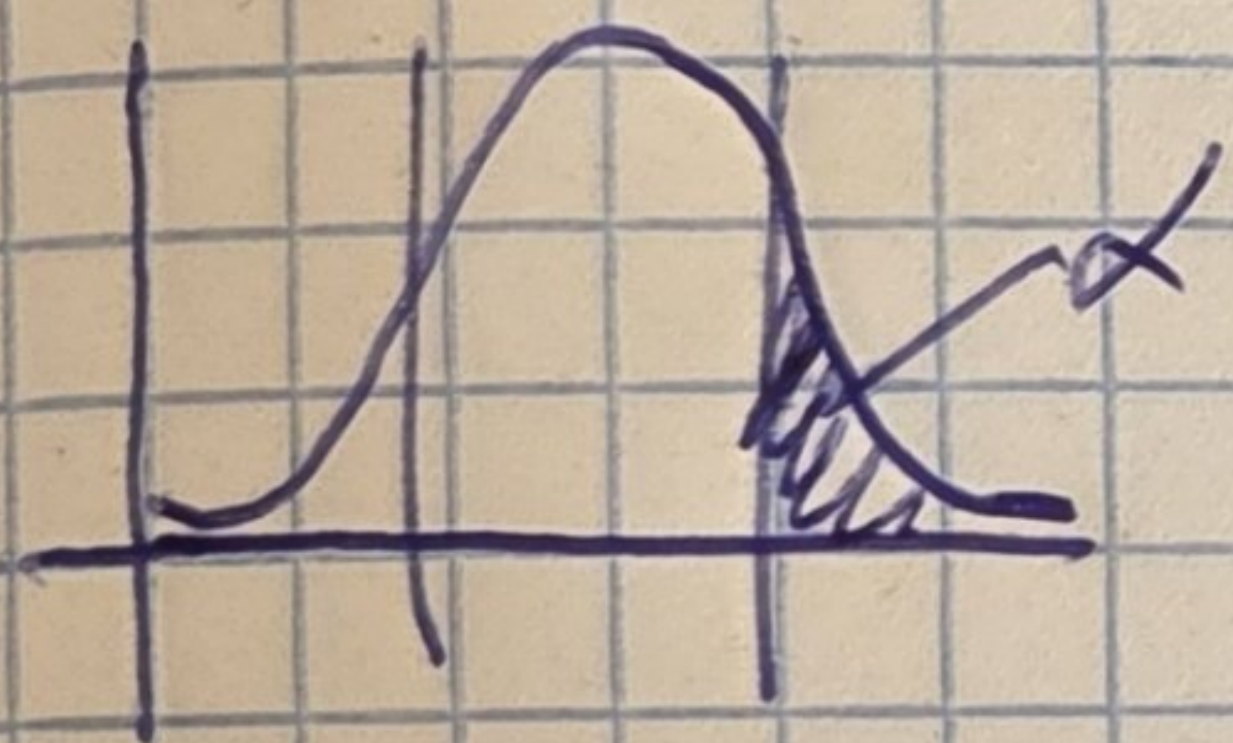
$$\bar{x} = \frac{-1 + 11 - 6 + 10 + 242}{3}$$

$$= 86$$

$$\bar{y} = \frac{-2.29 - 2.91}{2} = -2.6$$

$$z = \frac{(\bar{x} - \bar{y}) - (a - b)}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \sim N(0, 1)$$

$$z = \frac{(\bar{x} - \bar{y})}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} = \frac{1.003}{\sqrt{\frac{2}{3}}} = 0.93$$



$$t_{1-\alpha} = t_{0.95} \approx 1.645 \text{ (для норм. распр.)}$$

$$G: [1.645; +\infty)$$

$$z \notin G \Rightarrow \text{нет осн. отв. } H_0$$

$$\text{Пусть } a - b = \theta$$

$$W(\theta) = P(z \in G | H_1) = P(z \geq t_{1-\alpha} | \theta) \quad \Theta$$

$$\text{т.к. } \bar{x} - \bar{y} \sim N(\theta, \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m})$$

$$z \sim N\left(\frac{\theta}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}, 1\right)$$

$$\Theta: P\left(z - \frac{\theta}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}} \geq t_{1-\alpha} - \frac{\theta}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}\right) \Rightarrow$$

$$\Rightarrow W(\theta) = P\left(z \geq t_{1-\alpha} - \frac{\theta}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}\right) = 1 - \Phi\left(t_{1-\alpha} - \frac{\theta}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}\right)$$